

Estimated Annual Agricultural Pesticide Use Power County, Idaho 2014-2018

24 July 2026

Summary

Power County ranks **#2 of Idaho's 44 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of ID's five highest pesticide-use counties. Power County, on the Snake River Plain near American Falls, grows irrigated potatoes, wheat, alfalfa, and sugar beets.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Power County, Idaho** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” -[Pesticide National Synthesis Project webpage](#)

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”).
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound
OTH	Other
PYR	Pyrethrins/Pyrethroids

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Power County, ID, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	METAM POTASSIUM	DTC	1047870	474664	320859
2	METAM	DTC	489220	702022	672298
3	SULFURIC ACID	INO	203660	369325	369791
4	CHLOROPICRIN	OC	199930	87780	87780
5	DICHLOROPROPENE	OC	107885	164949	137116
6	GLYPHOSATE	OTH	39940	36419	34052
7	2,4-D	CPX	16103	10775	9056
8	CHLOROTHALONIL	OTH	14806	13503	11229
9	PENDIMETHALIN	OTH	12196	11460	9637
10	METRIBUZIN	TRZ	9265	9647	8226
11	MCPA	CPX	9180	7754	9464
12	EPTC	TC	8694	11177	10528
13	MALEIC HYDRAZIDE	OTH	7050	4574	4857
14	DIMETHENAMID & DIMETHENAMID-P	OTH	6861	7472	6025
15	DIMETHENAMID-P	OTH	6861	6662	5539
16	MANCOZEB	DTC	6129	5794	8513
17	METOLACHLOR & METOLACHLOR-S	OTH	5688	4622	3974
18	PHOSPHORIC ACID	INO	5543	5209	7350
19	METHOMYL	CB	5155	4269	3173
20	METOLACHLOR-S	OTH	5144	4016	3352
21	PARAQUAT	DIP	4348	4151	3089
22	BROMOXYNIL	OTH	4271	2853	3238
23	ACETOCHLOR	OTH	3983	1655	1638
24	FLUROXYPYR	OTH	3585	3994	3653
25	ATRAZINE	TRZ	2882	3886	2385
26	QUINTOZENE	OTH	2794	1553	1553
27	PYRIMETHANIL	OTH	2602	2172	1941
28	DIQUAT	DIP	2320	2163	2362
29	AZOXYSTROBIN	OTH	2026	1813	1780
30	PROPARGITE	OTH	1979	1979	2454
31	GLUFOSINATE	OTH	1897	951	706
32	BOSCALID	OTH	1870	1932	1796
33	CHLORPYRIFOS	OP	1804	2778	3118
34	IMIDACLOPRID	OTH	1543	1505	1556
35	PYRACLOSTROBIN	OTH	1543	1223	1306
36	FLUOPYRAM	OTH	1260	1021	795
37	SULFUR	INO	1187	5591	5180
38	PROPICONAZOLE	OTH	1038	1112	1074
39	PINOXADEN	OTH	1019	808	705
40	OXAMYL	CB	1012	928	2761
41	TRIFLURALIN	OTH	929	993	707
42	CLOPYRALID	OTH	765	1137	908
43	THIOPHANATE-METHYL	OTH	676	1169	854
44	DICAMBA	OTH	664	1143	1081
45	FLUTOLANIL	OTH	629	1562	1429
46	METOLACHLOR	OTH	544	606	622
47	FLUXAPYROXAD	OTH	524	376	298
48	PROTHIOCONAZOLE	OTH	521	252	163
49	TEBUCONAZOLE	OTH	498	592	372

Table 2: Top 80 EPest-high Estimates, Power County, ID, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) *(continued)*

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
50	MEFENOXAM	OTH	468	666	580
51	CARFENTRAZONE-ETHYL	OTH	460	328	216
52	CYFLUTHRIN	PYR	453	221	137
53	CYPROCONAZOLE	OTH	448	287	287
54	METIRAM	DTC	438	438	670
55	DIFENOCONAZOLE	OTH	433	369	492
56	FLUAZINAM	OTH	341	333	479
57	HEXYTHIAZOX	OTH	334	267	220
58	METALAXYL	OTH	287	287	529
59	PYROXASULFONE	OTH	269	151	431
60	MANDIPROPAMID	OTH	264	206	238
61	THIFENSULFURON	OTH	258	228	189
62	PYRASULFOTOLE	OTH	255	115	147
63	METCONAZOLE	OTH	253	347	277
64	SPIROTETRAMAT	OTH	227	186	114
65	DIMETHOATE	OP	226	2569	2426
66	PICLORAM	OTH	216	75	46
67	PICOXYSTROBIN	OTH	213	172	173
68	COPPER HYDROXIDE	INO	188	930	755
69	ZETA-CYPERMETHRIN	PYR	180	163	182
70	HEXAZINONE	TRZ	167	213	137
71	FLUCARBAZONE	OTH	166	97	60
72	LINURON	OTH	153	160	99
73	BIFENTHRIN	PYR	151	84	124
74	CYHALOTHRIN-LAMBDA	PYR	149	257	192
75	DIURON	OTH	140	209	166
76	HYDROGEN PEROXIDE	INO	121	121	121
77	COPPER OXYCHLORIDE	INO	102	249	228
78	CLOTHIANIDIN	OTH	101	101	131
79	PROSULFURON	OTH	99	95	100
80	TRIBENURON METHYL	OTH	94	99	85

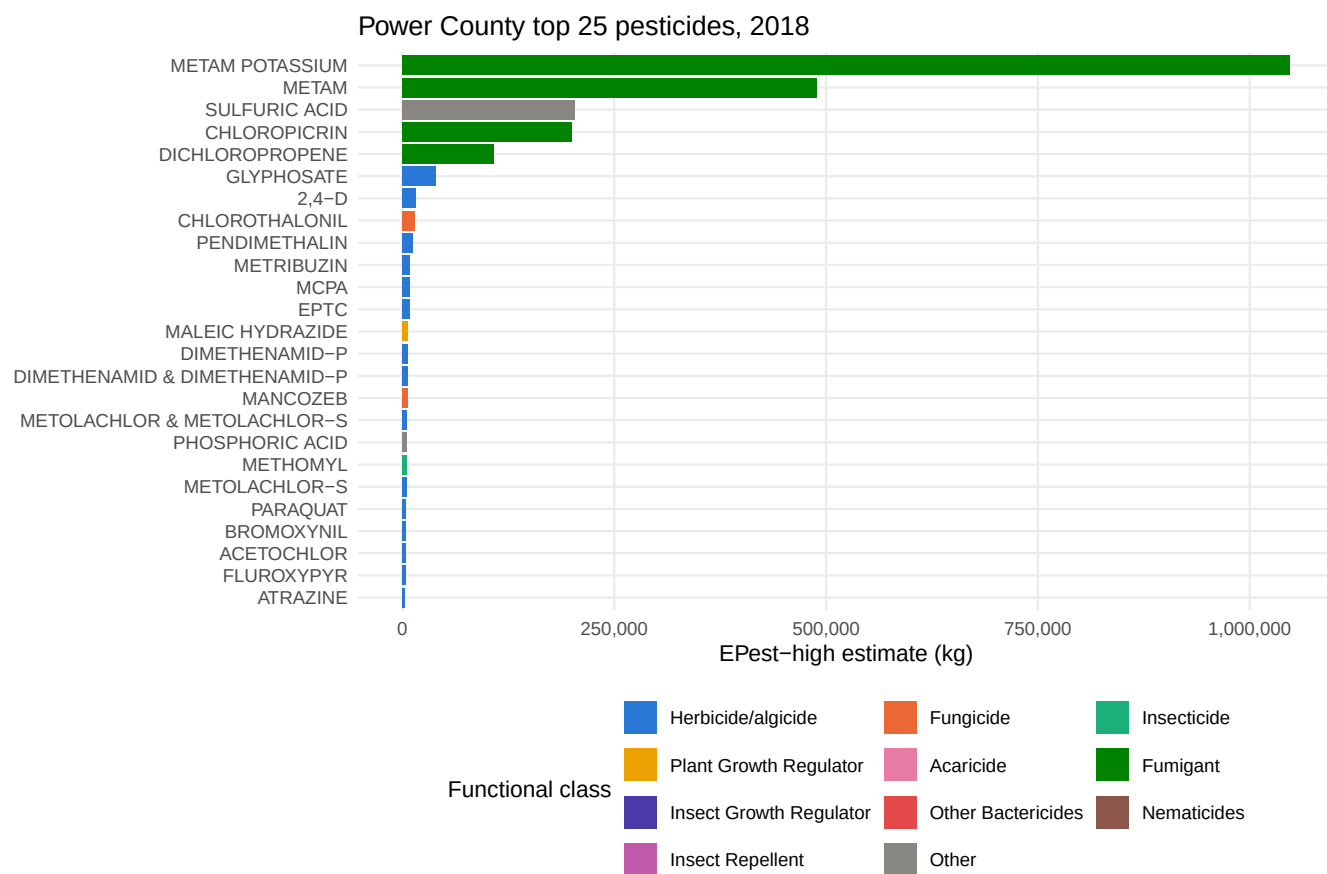


Figure 1: Top 25 compounds in Power County by Epest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Power County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM POTASSIUM	1047870	1047870
2	METAM	489220	489220
3	SULFURIC ACID	203660	203660
4	CHLOROPICRIN	0	199930
5	DICHLOROPROPENE	107885	107885
6	GLYPHOSATE	26388	39940
7	2,4-D	15923	16103
8	CHLOROTHALONIL	14806	14806
9	PENDIMETHALIN	11585	12196
10	METRIBUZIN	9196	9265
11	MCPA	9180	9180
12	EPTC	8694	8694
13	MALEIC HYDRAZIDE	0	7050
14	DIMETHENAMID & DIMETHENAMID-P	5641	6861
15	DIMETHENAMID-P	5641	6861
16	MANCOZEB	6129	6129
17	METOLACHLOR & METOLACHLOR-S	4696	5688
18	PHOSPHORIC ACID	5543	5543
19	METHOMYL	0	5155
20	METOLACHLOR-S	4276	5144
21	PARAQUAT	0	4348
22	BROMOXYNIL	4271	4271
23	ACETOCHLOR	0	3983
24	FLUROXYPYR	3550	3585
25	ATRAZINE	0	2882
26	QUINTOZENE	0	2794
27	PYRIMETHANIL	2602	2602
28	DIQUAT	2320	2320
29	AZOXYSTROBIN	2026	2026
30	PROPARGITE	0	1979
31	GLUFOSINATE	1897	1897
32	BOSCALID	1870	1870
33	CHLORPYRIFOS	1754	1804
34	IMIDACLOPRID	1543	1543
35	PYRACLOSTROBIN	1522	1543
36	FLUOPYRAM	1237	1260
37	SULFUR	1184	1187
38	PROPICONAZOLE	1038	1038
39	PINOXADEN	1019	1019
40	OXAMYL	1012	1012
41	TRIFLURALIN	76	929
42	CLOPYRALID	347	765
43	THIOPHANATE-METHYL	64	676
44	DICAMBA	597	664
45	FLUTOLANIL	629	629
46	METOLACHLOR	420	544
47	FLUXAPYROXAD	514	524
48	PROTHIOCONAZOLE	498	521
49	TEBUCONAZOLE	498	498
50	MEFENOXAM	468	468

Table 4: Top 50 Power County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	702022	702022
2	METAM POTASSIUM	474664	474664
3	SULFURIC ACID	369325	369325
4	DICHLOROPROPENE	164949	164949
5	CHLOROPICRIN	10524	87780
6	PETROLEUM OIL	830	68444
7	GLYPHOSATE	31881	36419
8	ETHOPROPHOS	0	32027
9	PROPAMOCARB HCL	0	28927
10	CHLOROTHALONIL	13503	13503
11	PENDIMETHALIN	10666	11460
12	EPTC	11032	11177
13	2,4-D	9954	10775
14	METRIBUZIN	9587	9647
15	MCPA	7744	7754
16	DIMETHENAMID & DIMETHENAMID-P	5857	7472
17	DIMETHENAMID-P	5857	6662
18	MANCOZEB	5794	5794
19	SULFUR	400	5591
20	PHOSPHORIC ACID	5209	5209
21	METOLACHLOR & METOLACHLOR-S	3555	4622
22	MALEIC HYDRAZIDE	0	4574
23	METHOMYL	0	4269
24	PARAQUAT	6	4151
25	METOLACHLOR-S	3401	4016
26	FLUROXYPYR	3971	3994
27	ATRAZINE	242	3886
28	BROMOXYNIL	2846	2853
29	CHLORPYRIFOS	2579	2778
30	DIMETHOATE	215	2569
31	DIMETHENAMID	0	2429
32	PYRIMETHANIL	2172	2172
33	DIQUAT	2163	2163
34	PROPARGITE	0	1979
35	BOSCALID	1932	1932
36	AZOXYSTROBIN	1813	1813
37	ACETOCHLOR	0	1655
38	FLUTOLANIL	1562	1562
39	QUINTOZENE	156	1553
40	IMIDACLOPRID	1504	1505
41	PYRACLOSTROBIN	1206	1223
42	THIOPHANATE-METHYL	22	1169
43	DICAMBA	720	1143
44	CLOPYRALID	131	1137
45	PROPICONAZOLE	1039	1112
46	SPIROMESIFEN	0	1072
47	FLUOPYRAM	1013	1021
48	TRIFLURALIN	163	993
49	GLUFOSINATE	951	951
50	COPPER HYDROXIDE	510	930

Table 5: Top 50 Power County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	672298	672298
2	SULFURIC ACID	369791	369791
3	METAM POTASSIUM	320859	320859
4	DICHLOROPROPENE	137116	137116
5	CHLOROPICRIN	10524	87780
6	PETROLEUM OIL	442	34253
7	GLYPHOSATE	31322	34052
8	PROPAMOCARB HCL	0	28927
9	ETHOPROPHOS	0	21719
10	CHLOROTHALONIL	11229	11229
11	EPTC	10337	10528
12	PENDIMETHALIN	9123	9637
13	MCPA	9445	9464
14	2,4-D	8553	9056
15	MANCOZEB	8513	8513
16	METRIBUZIN	8156	8226
17	PHOSPHORIC ACID	7350	7350
18	DIMETHENAMID & DIMETHENAMID-P	4623	6025
19	DIMETHENAMID-P	4623	5539
20	SULFUR	280	5180
21	MALEIC HYDRAZIDE	17	4857
22	METOLACHLOR & METOLACHLOR-S	2832	3974
23	FLUROXYPYR	3639	3653
24	METOLACHLOR-S	2728	3352
25	BROMOXYNIL	2995	3238
26	METHOMYL	0	3173
27	CHLORPYRIFOS	2871	3118
28	PARAQUAT	13	3089
29	OXAMYL	2761	2761
30	PROPARGITE	0	2454
31	DIMETHENAMID	0	2429
32	DIMETHOATE	133	2426
33	ATRAZINE	158	2385
34	DIQUAT	2362	2362
35	PYRIMETHANIL	1941	1941
36	BOSCALID	1796	1796
37	AZOXYSTROBIN	1780	1780
38	ACETOCHLOR	0	1638
39	IMIDACLOPRID	1556	1556
40	QUINTOZENE	156	1553
41	FLUTOLANIL	1429	1429
42	PYRACLOSTROBIN	1225	1306
43	DICAMBA	509	1081
44	PROPICONAZOLE	940	1074
45	CLOPYRALID	165	908
46	THIOPHANATE-METHYL	24	854
47	FLUOPYRAM	790	795
48	SODIUM CHLORATE	765	765
49	COPPER HYDROXIDE	503	755
50	TRIFLURALIN	132	707